

Mapping O*NET Tasks to ISCO Occupations using Text Similarity

Majid Einian*

Abstract

O*NET task data are widely used to measure technology exposure and skill demand, but O*NET is based on the U.S. Standard Occupational Classification (SOC) while international labour statistics use ISCO. Existing SOC–ISCO concordances operate at the occupation level, ignoring within-occupation task variation. This paper constructs a task-level crosswalk from O*NET to 4-digit ISCO-08 occupations using text similarity and ESCO information, assigning 18,796 tasks to 435 of 436 ISCO-08 groups. Validation against institutional crosswalks shows 84.6% exact agreement and 94.7% major-group agreement. The crosswalk enables task-based measures of automation, skills, and AI exposure to be applied to internationally comparable labour statistics.

Keywords: Occupational classification; O*NET; ISCO-08; Task-level crosswalk; Text similarity; Labour market

JEL codes: J24; J44; C88; O33; C80

1 Introduction

Many studies in labour economics use O*NET task data to measure technology exposure, skill demand, and occupational change (Autor et al., 2003; Acemoglu and Autor, 2011; Frey and Osborne, 2017; Acemoglu and Restrepo, 2020; Atalay et al., 2020). More recent datasets measuring exposure to artificial intelligence and digital technologies are also published at the O*NET task level (Eloundou et al., 2024; Shao et al., 2025; Handa et al., 2025; Appel, McCrory, et al., 2025; Appel, Massenkoff, et al., 2026; Massenkoff et al., 2026), based on both the SOC 2010 and SOC 2018 versions of O*NET.

O*NET is organised under the U.S. Standard Occupational Classification (SOC), while most international employment statistics are indexed using the International Standard Classification of Occupations (ISCO) (International Labour Office, 2012). Existing SOC–ISCO concordances operate at the occupation level (ESCO Secretariat, 2022; Bureau of Labor Statistics, 2015; Hardy et al., 2016). These mappings allow researchers to translate occupation-level measures across classification systems, but they ignore differences in tasks within occupations. Task-level information is central to much of the recent literature on technological change, where exposure to automation or artificial intelligence is measured using detailed descriptions of work

*Corresponding author. Arcada University of Applied Sciences, Jan-Magnus Janssons plats 1, 00560 Helsinki, Finland. Email: majid.einian@arcada.fi

activities rather than occupational titles alone. Aggregating tasks to occupations before applying a crosswalk removes this within-occupation variation and reduces the precision of empirical analysis.

This paper builds a direct task-level crosswalk from O*NET to 4-digit ISCO-08 occupations using text similarity. The method compares the text of O*NET task statements with occupational descriptions from ISCO-08 and ESCO, allowing each task to be matched to the most similar ISCO-08 occupation group. Two crosswalk versions are provided, corresponding to O*NET releases based on SOC 2018 and SOC 2010 classifications, allowing compatibility with both recent and earlier task-based datasets.

The resulting crosswalk assigns 18,796 O*NET tasks to 435 of 436 ISCO-08 unit groups. Validation against existing SOC–ISCO concordances shows 84.6% exact agreement and 94.7% agreement at the major-group level. The mapping is also highly stable across O*NET releases. Among 14,174 tasks shared between O*NET 29.2 (SOC 2018) and O*NET 25.0 (SOC 2010), assignments agree in 100% of cases. The crosswalk allows existing task-based measures of automation, skills, and AI exposure to be applied directly to internationally comparable labour statistics.

2 Data

2.1 Occupational data

The primary task data source is O*NET 29.2, the current release under the SOC 2018 taxonomy (Office of Management and Budget, 2018), which contains 18,796 task statements keyed by official task ID. Each statement describes a discrete work activity performed within a specific SOC occupation, and the parent occupation title is used as additional context in the matching procedure. A secondary release is built from O*NET 25.0, the final release under SOC 2010, which contains 19,735 task statements and enables application to datasets constructed using the earlier taxonomy, including early versions of the Anthropic Economic Index (Appel, McCrory, et al., 2025; Appel, Massenkoff, et al., 2026). O*NET is a standard source of occupational task information in the labour economics literature; see Peterson et al. (2001) for a detailed description.

The target classification is ISCO-08 at the four-digit unit-group level (International Labour Office, 2012). ISCO-08 is the international standard used in most cross-country labour statistics. Official ISCO-08 task descriptions are available for 435 of the 436 unit groups and form the main target-side text used in the matching procedure. The remaining group, ISCO-08 1113 (Traditional Chiefs and Heads of Village), has no clear counterpart in U.S. occupational data and is therefore not reachable from O*NET.

Data from ESCO are also used (European Commission, 2019). ESCO includes approximately 3,000 occupation concepts linked to ISCO-08 unit groups and adds skill and competence information that complements the official ISCO-08 task descriptions.

2.2 Institutional crosswalks for validation

The crosswalk is evaluated using four existing SOC–ISCO-08 concordances. These are used only for validation and do not affect the construction of the task-level mapping.

For SOC 2018, two concordances linking ESCO and O*NET occupations are used: the ESCO–SOC 2018 crosswalk (3,834 occupation pairs) (O*NET Center, 2022) and the reverse mapping, the SOC 2018–ESCO crosswalk (2,756 pairs) (ESCO Secretariat, 2022).

For SOC 2010, the official U.S. Bureau of Labor Statistics SOC 2010–ISCO-08 crosswalk (1,112 pairs) (Bureau of Labor Statistics, 2015) and a SOC 2010–ISCO-08 crosswalk published by the Institute for Structural Research (IBS), Warsaw (1,111 pairs) (Hardy et al., 2016) are used.

3 Method

3.1 Overview

The pipeline maps each O*NET task statement to a single ISCO-08 unit group using text similarity. Figure 1 summarises the data flow. Each task is represented using its text description and its parent SOC occupation title. Each ISCO-08 occupation is represented using official ISCO-08 task descriptions enriched with ESCO occupation information and associated skill texts.

All texts are encoded with the `all-mpnet-base-v2` sentence embeddings (Reimers and Gurevych, 2019). Cosine similarity between task and occupation representations determines matches. The assignment procedure selects the most similar ISCO-08 group for each task subject to minimum similarity and coverage constraints.

The approach preserves task-level variation within occupations and allows task-based measures derived from O*NET to be translated to internationally comparable labour statistics.

3.2 Task and occupation representations

Each O*NET task is represented using the task statement text, the title of the SOC occupation to which the task belongs, and optionally the associated Detailed Work Activity (DWA) labels. All text components are encoded by the `all-mpnet-base-v2` sentence encoder into unit-normalised real-valued vectors: each embedding $e \in \mathbb{R}^{768}$, $\|e\|_2 = 1$. All mixing weights are scalars $w \in [0, 1]$, so every composite representation is a convex combination of unit vectors and is itself a vector in $\mathbb{R}^{768}$. Task embeddings are combined as

$$q = (1 - w_{\text{soc}})[w_{\text{dwa}} e_{\text{DWA}} + (1 - w_{\text{dwa}}) e_{\text{task}}] + w_{\text{soc}} e_{\text{SOC}}, \quad (1)$$

where e_{task} is the embedding of the task text, e_{DWA} is the embedding of the associated DWA labels, and e_{SOC} is the embedding of the SOC occupation title. Including occupation titles helps distinguish tasks that use similar language but occur in different occupational contexts.

Each ISCO-08 occupation is represented using both official ISCO-08 task descriptions and ESCO occupation information. The ISCO component combines task descriptions with summary

occupation text,

$$e_{\text{ISCO}} = w_{\text{isco,task}} e_{\text{tasks}} + (1 - w_{\text{isco,task}}) e_{\text{info}}. \quad (2)$$

The ESCO component combines occupation descriptions with associated skill text,

$$e_{\text{ESCO}} = w_{\text{occ}} e_{\text{occ}} + (1 - w_{\text{occ}}) e_{\text{skills}}. \quad (3)$$

Where multiple ESCO occupations map to the same ISCO-08 unit group, their embeddings are averaged.

The final occupation representation is a weighted combination of ISCO and ESCO information,

$$e_{\text{target}} = w_{\text{isco}} e_{\text{ISCO}} + (1 - w_{\text{isco}}) e_{\text{ESCO}}. \quad (4)$$

Combining ISCO and ESCO information improves coverage and provides richer context on occupational tasks and required skills.

3.3 Assignment rule

For each task representation q , cosine similarity $\text{sim}(q, e_{\text{target}}) = q \cdot e_{\text{target}} \in [-1, 1]$ (equivalent to the dot product because both vectors are unit-normalised) is computed against all ISCO-08 occupation representations. Each task is assigned to the occupation with the highest similarity score when that score exceeds a minimum threshold and the difference between the best and second-best match exceeds a margin threshold.

A coverage constraint ensures that ISCO-08 unit groups with relevant task content are represented in the crosswalk when similarity levels are sufficiently high. This supports aggregation of task-level measures to occupation-level statistics.

Parameter values are selected using an unsupervised search procedure described in Section 5.

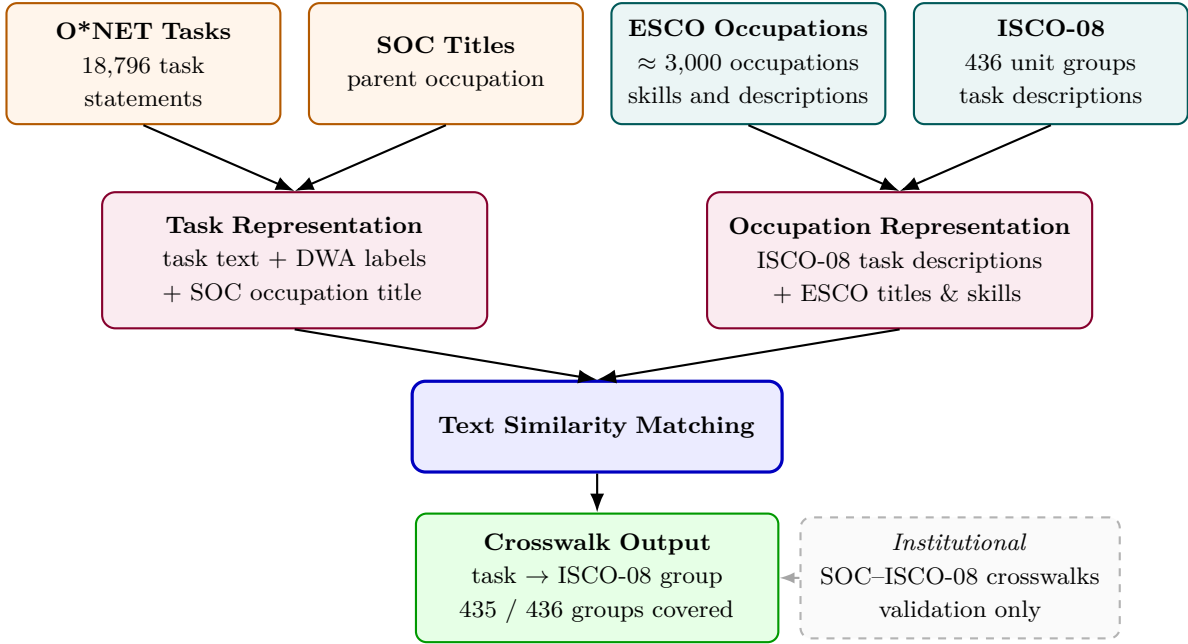


Figure 1: Task-level mapping from O*NET to ISCO-08.

Notes: Task representations combine O*NET task text, DWA labels, and SOC occupation title. Occupation representations combine official ISCO-08 task descriptions with ESCO occupation titles and skills. Similarity matching assigns each task to one ISCO-08 unit group. Institutional SOC-ISCO-08 crosswalks are used only for validation.

4 Validation

Because no gold standard exists for mapping tasks directly to ISCO-08 occupations, validation is performed indirectly using institutional occupation-level SOC-ISCO crosswalks and additional qualitative assessment. The task-level mapping is first aggregated to the occupation level, allowing comparison with existing concordances constructed by statistical agencies and international organisations. These crosswalks provide an external benchmark for evaluating whether semantic similarity between task and occupation text produces assignments consistent with established classification correspondences.

Agreement with institutional SOC-ISCO crosswalks is high. For the O*NET 29.2 (SOC 2018) version, 84.6% of SOC occupations are mapped to the same ISCO-08 unit group as in at least one institutional crosswalk, while 94.7% agree at the ISCO major-group level.

Among the 14,174 tasks shared between O*NET 29.2 (SOC 2018) and O*NET 25.0 (SOC 2010), assignments agree in 100% of cases.

Agreement rates are robust to the choice of reference crosswalk; detailed tables are in Online Appendix C.

Results are stable across parameter choices and across O*NET releases. Sensitivity analysis varying embedding models, component weights, and similarity thresholds produces similar agreement rates, indicating that the mapping is not driven by specific parameter settings. Online Appendix A reports agreement statistics across alternative specifications. The version constructed using O*NET 25.0 (SOC 2010) produces very similar results, supporting the use of

the crosswalk with datasets based on earlier O*NET releases, including early versions of the Anthropic Economic Index.

Qualitative inspection confirms semantic consistency; examples and mismatches are detailed in Online Appendix E.

5 Parameter Selection

The main specification uses $w_{\text{dwa}} = 0.125$, $w_{\text{soc}} = 0.25$, $w_{\text{isco}} = 0.5$, $w_{\text{isco,task}} = 0.8125$, and $w_{\text{occ}} = 0.125$. The similarity threshold is set to 0.45 with margin parameter 0.03. Each task is assigned to the single most similar ISCO-08 unit group.

Parameter values are chosen using unsupervised criteria reflecting the objectives of the mapping. Candidate specifications are evaluated based on coverage of ISCO-08 unit groups, mean similarity between matched texts, and the distribution of tasks across occupation groups. Configurations that produce excessive task concentration in broad ISCO groups are penalised; such concentration arises at extreme parameter values and reflects poor semantic differentiation rather than structural taxonomy differences. The selected configuration’s low concentration share (2.9%) reflects structural SOC–ISCO-08 differences: the two groups above the threshold each draw over 75% of their tasks from a single SOC major group (Online Appendix E).

Three qualitative patterns emerge. First, a small DWA weight ($w_{\text{dwa}} = 0.125$) improves performance over excluding DWA entirely, but large weights reduce it, suggesting that DWA text provides limited supplementary information beyond the task statement. Second, a moderate SOC occupation title weight ($w_{\text{soc}} = 0.25$) produces more stable assignments than heavier reliance on either component alone. Third, combining ISCO-08 task descriptions with ESCO occupation text improves retrieval relative to using either source separately, indicating that the two sources provide complementary information.

The selected specification assigns tasks to 435 of the 436 ISCO-08 unit groups, with ISCO 1113 structurally unmatched due to the absence of a functional equivalent in O*NET. Sensitivity analysis reported in Online Appendix A shows that results are stable across reasonable variation in parameter values.

6 Conclusion

This paper presents a semantic crosswalk that maps 18,796 task statements from O*NET 29.2 to ISCO-08 four-digit unit groups, covering 435 of the 436 groups in the classification. The approach combines task text with occupational context and enriched ISCO-08 representations, producing 84.6% exact agreement with institutional SOC–ISCO concordances and consistent results across O*NET releases. The crosswalk allows task-based measures derived from O*NET—including indices of automation exposure, skill content, and recent LLM-based occupational assessments—to be applied directly to internationally comparable labour statistics organised under ISCO-08. This enables cross-country analysis of technological change at the task level without requiring occupation-level aggregation prior to translation between classification systems. One limitation is structural: because O*NET reflects the U.S. occupational structure, ISCO

groups without a functional U.S. counterpart cannot be matched directly. In practice, this affects a small number of cases, most notably ISCO 1113 (Traditional Chiefs and Heads of Village). The crosswalk dataset and replication materials are provided in the Online Appendix and at <https://doi.org/10.5281/zenodo.20359118>.

Acknowledgements

Funding from Stiftelsen Arcada (Application No. 419) is gratefully acknowledged.

Data Availability

The crosswalk dataset and replication materials are available at <https://doi.org/10.5281/zenodo.20359118>.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic), ChatGPT (OpenAI), Claude Code (Anthropic), and Codex (OpenAI) for language editing and writing assistance, and for code generation in the development of the text similarity pipeline. After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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